

PHYS 427: Lecture # 4

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Lecture Span

1. Entropy of an ideal gas

Review

Let f has a temperature T and the reservoir R has another temperature T_1 . We assume that $R \gg f$. The reservoir exchanges heat with the system f , until both have the same temperature ($T = T_1$).

Paramagnets

The entropy is a negative parabola shape. Here, then

$$\frac{1}{T} = \frac{\partial S}{\partial U} \quad (1)$$

On the LHS of the graph, we have that the temperature is negative. Vice versa on the RHS of the graph. Suppose you put the paramagnet with $T = \infty$ in contact with another system at finite temperature T_B . The heat should flow from the hotter to the colder system. Calculate

$$\Delta S_{tot} = \Delta S_A + \Delta S_B \quad (2)$$

$$= \frac{\partial S_A}{\partial U_A} \Delta U_A + \frac{\partial S_B}{\partial U_B} \Delta U_B \quad (3)$$

$$= \left(\frac{1}{T_A} - \frac{1}{T_B} \right) \Delta U_A \quad (4)$$

$$= \left(-\frac{1}{T_B} \right) \Delta U_A \geq 0 \quad (5)$$

Therefore, ΔU_A must be negative, which means that energy flows from U_A into U_B , as expected. now, if $T_A < 0$ and $T_B > 0$, then

$$\Delta S_{tot} = \Delta S_A + \Delta S_B \quad (6)$$

$$= \frac{\partial S_A}{\partial U_A} \Delta U_A + \frac{\partial S_B}{\partial U_B} \Delta U_B \quad (7)$$

$$= \left(\frac{1}{T_A} - \frac{1}{T_B} \right) \Delta U_A \quad (8)$$

$$= \left(-\frac{1}{|T_A|} - \frac{1}{|T_B|} \right) \Delta U_A \quad (9)$$

Here, it shows that energy flows from A to B . This means that the system with $T < 0$ behaves like a system that is hotter than the other system with $T > 0$.

Entropy of an Ideal Gas

Consider free, independent, indistinguishable particles. Assume monatomic gas. Consider a volume with dimensions L_x, L_y, L_z . Then the volume of the box is $V = L_x L_y L_z$.

Now, we consider the multiplicity $\Omega(U, N)$ of this configuration. From quantum mechanics, recall that

$$H\Psi = \epsilon\Psi(r, t) \quad (10)$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi(r, t) = \epsilon\psi(r, t) \quad (11)$$

Then

$$\epsilon = \frac{p^2}{2m} = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2) \quad (12)$$

Where

$$k_i = \frac{\pi n_i}{L_i} \quad (13)$$

Where $n_i = 1, 2, 3, \dots$ are the quantum numbers. The energy is quantized, as

$$\epsilon = \frac{\hbar^2\pi^2}{2m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right) \quad (14)$$

Then, we wish to find $\Omega(U, N)$, i.e., finding all the microstates that have the same energy. Or the same number of integers $\{n_x, n_y, n_z\}$ with the same energies.

Working in the k space, we can draw a square around each point in k (on the k_x - k_y plane) with a x length of π/L_x and a y length of π/L_y . Then,

$$|k| = \sqrt{k_x^2 + k_y^2 + k_z^2} \quad (15)$$

Then, if we construct a sphere, then Ω is the # of lattice points which intersect the surface of this sphere. There is no simple, exact solution for this problem. But we can ask how many states are within energy less of ϵ . How many microstates have an energy $\leq \epsilon$.

If we want to do this, we can just calculate the volume and divide by the volume of each sphere. Each state occupies a box in the k space with a volume

$$\frac{\pi^3}{L_x L_y L_z} = \frac{\pi^3}{V} \quad (16)$$

Therefore, the # of microstates with energy $\leq \epsilon$ is approximately

$$\approx \frac{\text{volume of sphere in } k \text{ space with radius } \sqrt{2m\epsilon/\hbar^2}}{\pi^3/V} \quad (17)$$

Then let $\Phi(\epsilon)$ be the # of microstates with energy $\leq \epsilon$. Then

$$\Phi(\epsilon) \approx \frac{\frac{4\pi}{3}k^3/8}{\pi^3/V} \quad (18)$$

$$= \frac{V}{6\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \epsilon^{3/2} \quad (19)$$

We only consider $1/8$ of the coordinate space since $k_i \geq 0$. This is a good approximation for $k \gg \Delta k = \pi/L$ and $\epsilon \gg \Delta\epsilon = \pi^2\hbar^2/(2mV^{2/3})$.

Consider # of microstates in a narrow range of energies within $[\epsilon, \epsilon + \delta\epsilon]$. Then the number of energies is approximately

$$\Omega(\epsilon, \epsilon + \delta\epsilon) = \Phi(\epsilon + \delta\epsilon) - \Phi(\epsilon) \approx \frac{\partial\Phi}{\partial\epsilon} \delta\epsilon \quad (20)$$

$$= \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{\epsilon} \delta\epsilon \quad (21)$$

2D ideal gas

We can also calculate $\Phi(\epsilon)$ in two dimensions. Then, each state occupies $\pi^2/L_x L_y = \pi^2/A$.

$$\Phi(\epsilon) = \frac{\text{area of circle}}{\text{area per state}} \quad (22)$$

$$= \frac{\pi k^2/4}{\pi^2/A} \quad (23)$$

$$= \frac{A}{4\pi} \left(\frac{2m}{\hbar^2} \right) \epsilon \quad (24)$$

We can get our multiplicity in two dimensions, with

$$\Omega(\epsilon, \epsilon + \delta\epsilon) = \frac{\partial\Phi}{\partial\epsilon} \delta\epsilon \quad (25)$$

$$= \frac{A}{4\pi} \left(\frac{2m}{\hbar^2} \right) \delta\epsilon \quad (26)$$

Density of states

$D(\epsilon)$ is the number of states per energy interval. Here,

$$D(\epsilon) = \frac{\partial\Phi}{\partial\epsilon} \quad (27)$$

Then

$$\Phi(\epsilon) = \int_0^\epsilon d\epsilon' D(\epsilon') \quad (28)$$

Then

$$\Omega(\epsilon, \epsilon + \delta\epsilon) = D(\epsilon) \delta\epsilon \quad (29)$$

However, everything we have done is for one particle in a box. How do we generalize to 10^{23} particles?

Extension to 2,3, ..., N particles

With 2 particles, you have a two particle wavefunction. Here, then

$$U = \epsilon_1 + \epsilon_2 = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} = \frac{\hbar^2}{2m} \left(\sqrt{\sum_{i,j} k_{ij}^2} \right) \quad (30)$$

Here, i_j is the i-th coordinate axis (x,y,z) for the j-th particle. Then, each two particle state occupies a space of

$$\left(\frac{\pi^3}{V} \right)^2 \quad (31)$$

You can generalize towards higher dimensional states. States with energy $\leq U$ occupy a volume of

$$\frac{1}{(2^3)^2} \text{ of the hypersphere} \quad (32)$$

For N particles, the states are on a lattice in $3N$ dimensional k space, with a volume of

$$\left(\frac{\pi^3}{V}\right)^N \quad (33)$$

States with energy less than U occupy a volume of $1/2^{3N}$ of the hypersphere in $3N$ dimensional k space. Then, $[U, U + \delta\epsilon]$ occupies volume of $1/2^{3N}$ of hypershell in $3N$ dim k space.

Here, $k \sim \sqrt{U}$ since $\epsilon = \hbar^2 k^2 / 2m \implies k \sim \sqrt{\epsilon}$.

1. ($N = 1$) $\Phi_1(U) \sim V k^3 \sim V U^{3/2}$
2. ($N = 2$) $\Phi_2(U) \sim V^2 k^6 \sim V^2 U^3$
3. (N) $\Phi_N(U) \sim V^N k^{3N} \sim V^N U^{3N/2}$

This implies that

$$\Omega_N(U, U + \delta U) \sim V^N U^{3N/2-1} \quad (34)$$

Avoid overcounting of the states, since the particles are indistinguishable. Then, you have to divide the multiplicity by $N!$. Then

$$\Omega(U, U + \delta U) = \frac{g_{3N}}{2^{3N} N!} \left(\frac{V}{\pi^3}\right)^N \left(\frac{2m}{\hbar^2}\right)^{3N/2} \frac{U^{3N/2-1}}{2} \delta U \quad (35)$$

Here, g_D is the solid angle of a hypershell in D dimensions. For instance, $g_3 = 4\pi$. We can also rewrite Ω as

$$\Omega_N(U, U + \delta U) = \frac{g_{3N}}{2^{3N+1} N!} \left(\frac{U}{\Delta\epsilon}\right)^{3N/2-1} \frac{\delta U}{\Delta\epsilon}, \quad \Delta\epsilon = \frac{\hbar^2 \pi^2}{2m V^{2/3}} \quad (36)$$

We then derive the entropy,

$$S(U, U + \delta U, V) = k_B \ln(\Omega_N) \quad (37)$$

$$= N k_B \ln V + \left(\frac{3N}{2} - 1\right) k_B \ln U + \text{other terms that don't depend on } U \text{ or } V \quad (38)$$

Then,

$$S(U, V) = N k_B \ln V + \frac{3N}{2} k_B \ln U + k_B \ln f(N) \quad (39)$$

what is f??? Then,

$$\frac{1}{T} = \frac{\partial S}{\partial U} = \frac{3N k_B}{2U} \quad (40)$$

Therefore, it follows that

$$U = \frac{3}{2} N k_B T \quad (41)$$

Define pressure

$$p = T \left(\frac{\partial S}{\partial V} \right)_{U, N} \quad (42)$$

An example: Consider then $V_{tot} = V_a + V_b$ is fixed. But V_i can vary. Then

$$S = S_A + S_B \quad (43)$$

Then

$$\frac{\partial S}{\partial V} = 0 = \frac{\partial S_A}{\partial V_A} \delta V_A + \frac{\partial S_B}{\partial V_B} \delta V_B \quad (44)$$

$$\frac{\partial S_A}{\partial V_A} = \frac{\partial S_B}{\partial V_B} \quad (45)$$

This is equilibrium. We can use the above equation to obtain the pressure and

$$PV = Nk_B T \quad (46)$$